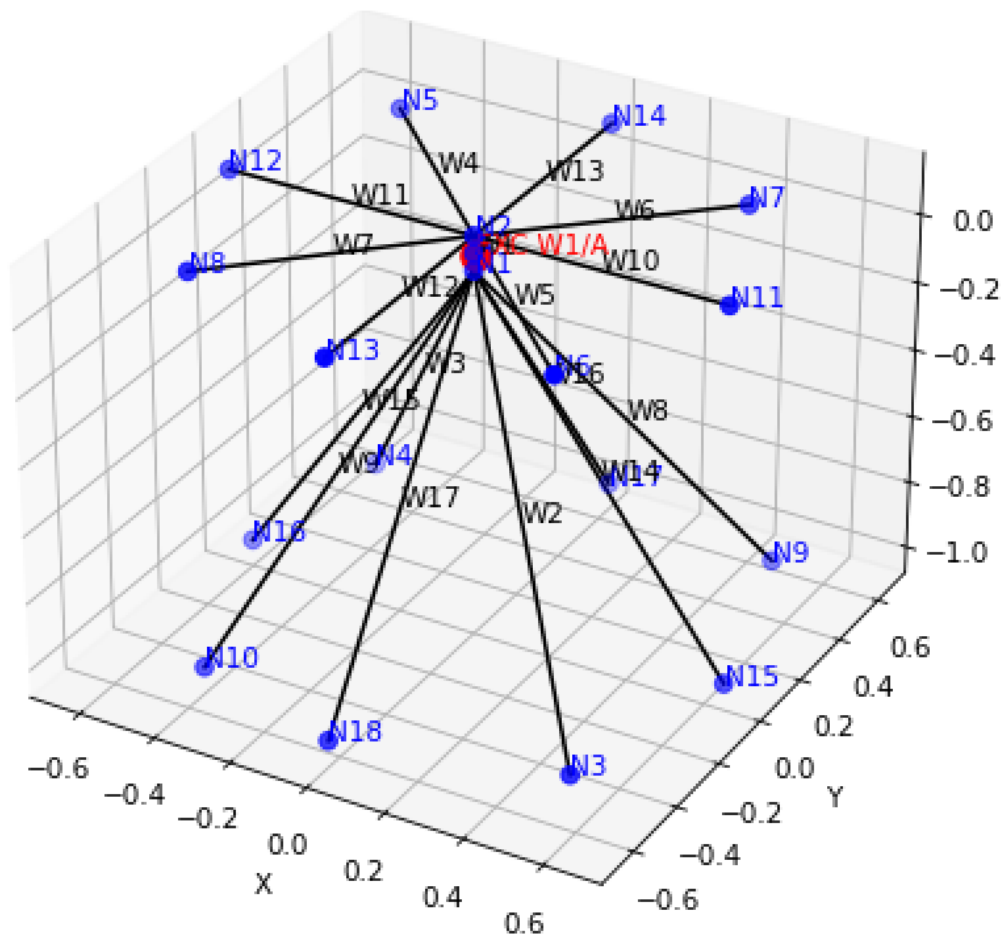


NEOMOM

“An amateur radio antenna prediction code”

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2026

Comments on modeling. From Lee

Abstract

NeoMoM is a wire-antenna modeling suite designed for amateur radio operators and antenna engineers. It predicts input impedances, SWR, gain and radiation patterns using a full-wave electromagnetic solver based on Maxwell's equations.

System Overview

The suite consists of three integrated applications that together support the complete antenna modeling workflow:

- **neomom_input** — Antenna geometry creation and editing
- **neomom** — Computational *Method of Moments* (MoM) engine
- **neomom_plot** — Visualization and interactive pattern analysis with geographic map overlay

Key Features

- **Open-source** design intended for community use, extension, and experimentation
- Modern Python-based graphical user interfaces for input and visualization
- Direct import of MMANA-GAL (*.maa) files with minor editing for excitation
- Export capability for NEC5 and MMANA-GAL input formats ready for further editing

Design Foundations

NeoMoM builds on the legacy of tools such as NEC, EZNEC, MiniNEC, and MMANA-GAL while introducing:

- A more rigorous computational framework
- A modern object-oriented architecture
- Improved usability through graphical interfaces

Advances in computing—such as increased memory, multi-core processors, and modern programming techniques—enable higher-performance simulations using object-oriented Fortran and optimized numerical libraries. AI now allows development of modern GUI applications.

Core Concepts

Node-Based Geometry Definition

Antenna structures are defined using a list of tagged spatial nodes (x, y, z coordinates). From these nodes:

- Wire elements are constructed as tagged polylines
- Excitation is defined at a specific wire and node
- Geometry can be created either via GUI or text-based input

Full-Sphere Radiation Modeling

Neomom computes radiation fields (E_θ and E_ϕ) over the entire sphere. This global dataset enables:

- Arbitrary azimuth and elevation pattern cuts
- Full 3D radiation visualization
- Geographic map-based pattern overlays

Extensible GUI Framework

The input and visualization interfaces are implemented in Python using AI-assisted development. This approach encourages:

- Custom GUI development by users
- Experimentation and adaptation within the amateur radio community

System Components

Neomom_input

- Creates and edits *.nml input files
- Provides real-time 3D visualization of antenna geometry as nodes, wires, and excitations are defined.
- Supports:
 - Global parameters (frequency, ground type, title)
 - Node coordinate input (user-defined units)
 - Wire and excitation definitions
- Reads existing *.nml* and *MMANA-GAL* (.maa) files

- Exports NEC and MMANA-GAL *.nec and *.maa files which, after editing, can be used in NEC5 (EZ-NEC) and MMANA-GAL.

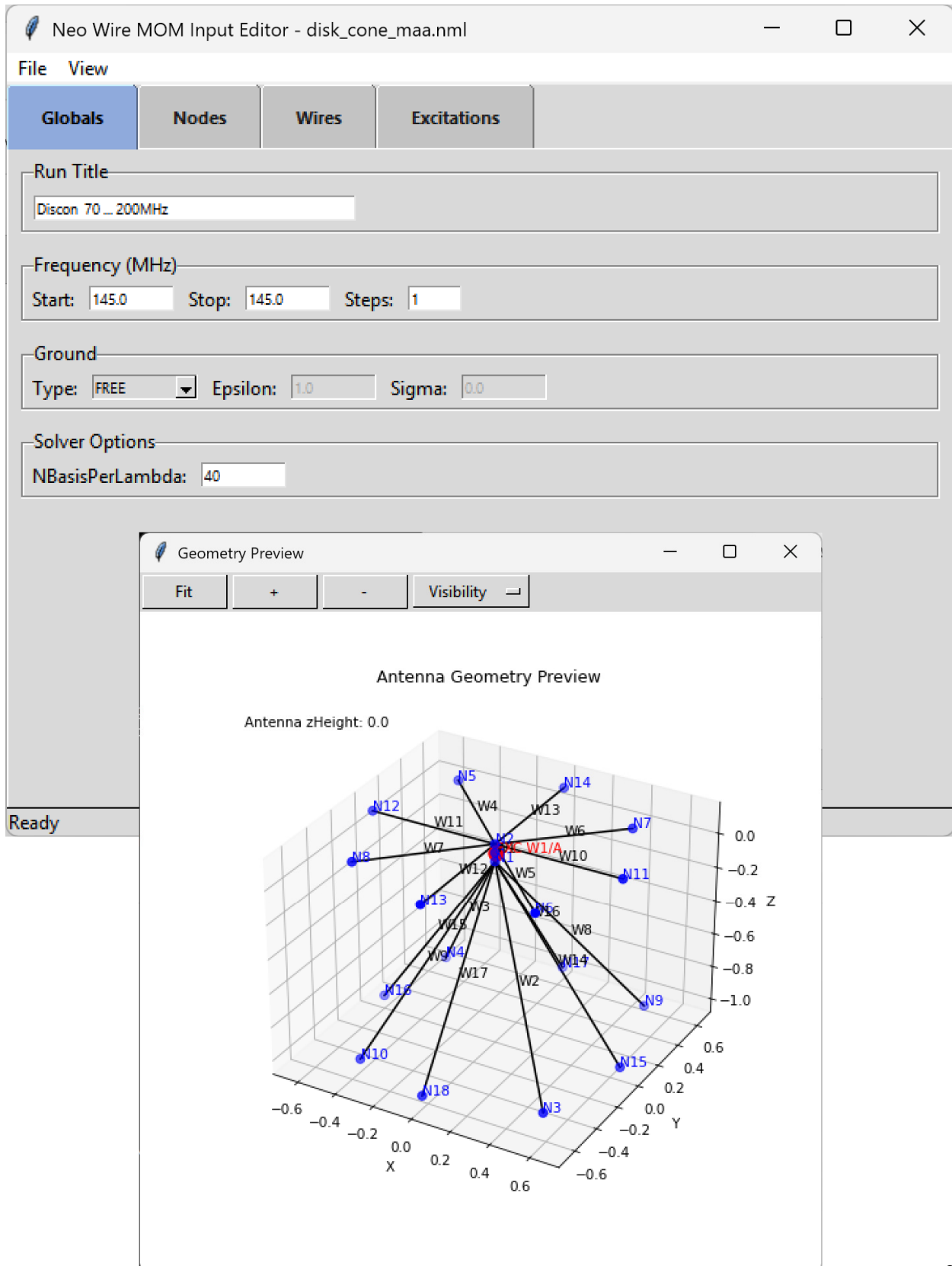
Neomom (Computational Engine)

- Full-wave Maxwell solver using the Method of Moments
- Triangle basis functions with Galerkin formulation
- Symmetric system matrix
- Input: Fortran namelist (*.nml)
- Output: Full-sphere complex E_θ and E_ϕ fields in *.csv format

Neomom_plot

- Input: Sphere values of E_θ and E_ϕ over full or half sphere, $[\theta, \phi]$, data as output by neomom
- Visualization options:
 - 2D azimuth cuts (user-selected elevation)
 - 2D elevation cuts (user-selected azimuth)
 - 3D radiation patterns with ground plane
 - Top-down geographic view aligned to compass directions
- Advanced features:
 - Radiation pattern overlay on world, CONUS, or regional maps
 - Multiple map projections (azimuthal equidistant, Lambert conformal, plate carrée)
 - Adjustable orientation relative to geographic north
 - Scaling options: ARRL, dBi, dBpeak, or linear gain (E_θ , E_ϕ , total)

neomom_input



neomom_plot

Antenna Pattern Viewer

Antenna

40 m Delta Loop at 10 m 29 MHz Peak gain: 13.5 dBi

$Z = 319.58 + j512.40 \, \Omega$ $SWR = 22.9$

Ground: REAL Height: 13 m (1.258 λ)

Visualization Mode

☒ 2D – Azimuth

☐ 2D – Elevation

☐ 3D – Standard

☐ 3D – Top-Down

☐ Map Overlay

2D Azimuth Options

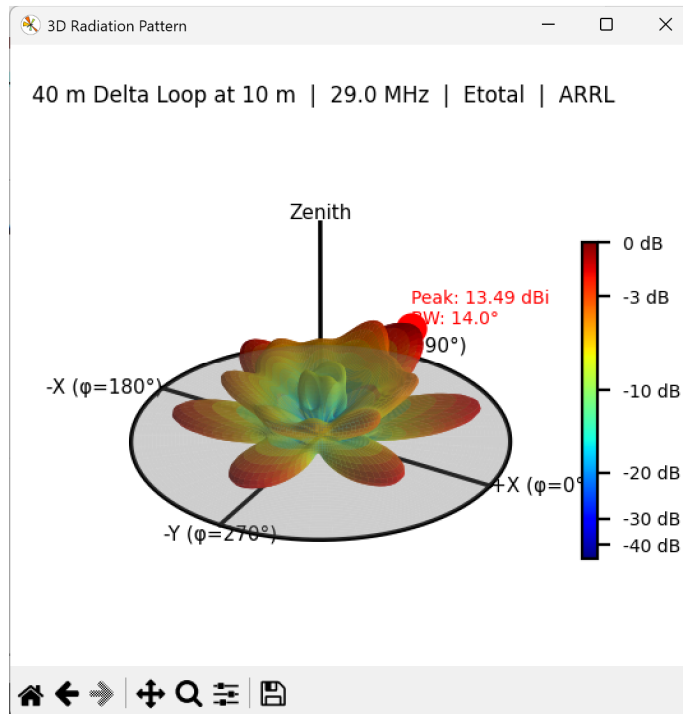
Components: Ev Eh Etotal

Scale: ARRL dB peak dBi Linear

theta = 80.0

Plot Azimuth

Ready.



Validation

The following tables show input impedances and gain computed with neomom, NEC5, and mmana-gal for:

- Half wave dipole, center fed
- Folded dipole, center fed
- End fed half wave dipole
- Quarter wave vertical at and near ground, $z = 0$
- Disk cone VHF array of top hat radials and sloped cone radials
- Yagi, 6m, six element array
- Delta loop sized for 40 m. Computed results for 40 and 10 m
- J Pole, 2m, above real ground

Dipole, folded dipole, and EFHW results are for free-space, perfect PEC ground, and real ground.

Half wave dipole, centre fed

$L = 20 \text{ m}$, $f = 7.3 \text{ MHz}$, wire radius = 1 mm

Free Space, Perfect Ground, Real Ground

FREE SPACE			
	Zin (ohms)	Gain (max) dB	
Neomom	71.9 - j 0.12	2.14	
Nec 5	71.6 - j 5.41	2.14	GE = 0, GN = -1
Mmana-gal	74.5 + j 2.05	1.99	

10 m above perfect ground			
	Zin (ohms)	Gain (max) dB	
Neomom	84.5 + j 30.5	7.58	
Nec 5	84.2 + j 25.0	7.58	GE = 1, GN = 1
Mmana-gal	74.1 + j 14.7	8.02	

5 m above perfect ground			
	Zin (ohms)	Gain (max) dB	
Neomom	31.1 + j 29.3	8.72	
Nec 5	31.6 + j 23.7	8.72	GE = 1, GN = 1
Mmana-gal	59.6 + j 2.6	5.0	

10 m above real ground (14.0,0.005)			
	Zin (ohms)	Gain (max) dB	
Neomom	83.1 + j 17.9	6.0	
Nec 5	82.5 + j 10.9	5.92	GE = 1, GN = 2
Mmana-gal	84.7 + j 32.6	5.87	

5 m above real ground (14.0,0.005)			
	Zin (ohms)	Gain (max) dB	
Neomom	48.4 + j 23.2	5.86	
Nec 5	55.6 + j 14.2	5.31	GE = 1, GN = 2
Mmana-gal	31.2 + j 31.3	7.88	

Folded dipole, centre fed

$L = 10.05 \text{ m}$, $f = 14.05 \text{ MHz}$, wire radius = 1.5 mm, $d = 0.2 \text{ m}$

fd20.maa file in the MMANA-GAL antenna library

Free Space, Perfect Ground, Real Ground

FREE SPACE			
	Zin (ohms)	Gain (max) dB	
Neomom	285 - j 0.74	2.19	
Nec 5	286 - j 10.4	2.14	GE = 0, GN = -1
Mmana-gal	286 + j 4.00	2.13	

5 m above PEC			
	Zin (ohms)	Gain (max) dB	
Neomom	328 + j 131	7.71	
Nec 5	333 + j 121	7.71	GE = 1, GN = 1
Mmana-gal	327 + j 136	7.7	

5 m above real gnd			
	Zin (ohms)	Gain (max) dB	
Neomom	320 + j 74.2	5.87	
Nec 5	321 + j 56.8	5.85	GE = 1, GN = 2
Mmana-gal	327 + j 136	5.83	

EFHF: End Fed Half Wave Dipole

$L = 20 \text{ m}$, $f = 7.3 \text{ MHz}$, wire radius = 1 mm

Free Space, Perfect Ground, Real Ground

FREE SPACE			
	Zin (ohms)	Gain (max) dB	
Neomom	1453 - j 1109	2.15	
Nec 5	1484 - j 1296	2.15	GE = 0, GN = -1
Mmana-gal	2036 - j 17481	2.17	

10 m above PEC			
	Zin (ohms)	Gain (max) dB	
Neomom	2143 - j 1400	7.6	
Nec 5	2196 - j 1578	7.6	GE = 1, GN = 1
Mmana-gal	1612 - j 18123	7.6	

10 m above real gnd			
	Zin (ohms)	Gain (max) dB	
Neomom	1853 - j 1315	6.0	
Nec 5	1862 - j 1486	5.9	GE = 1, GN = 2
Mmana-gal	1612 - j 18123	5.9	

Quarter Wave Vertical

$L = 10 \text{ m}$, $f = 7.3 \text{ MHz}$, wire radius = 1 mm

Perfect Ground, Real Ground

$z_H = 0.0$ and 0.1 m

PEC Ground

0.0 m above PEC			
	Zin (ohms)	Gain (max) dB	
Neomom	27.2 - j 2334	5.18	
Nec 5	36.7 - j 2.5	5.15	GE = 1, GN = 1
Mmana-gal	10.5 - j 17379	8.05	

0.1 m above PEC			
	Zin (ohms)	Gain (max) dB	
Neomom	26.9 - j 2421	5.19	
Nec 5	28.0 - j 2781	5.19	GE = 1, GN = 1
Mmana-gal	10.5 - j 17379	8.04	

Real Ground

0.0 m above real gnd			
	Zin (ohms)	Gain (max) dB	
Neomom	47.0 - j 2429	-2.5	
Nec 5	48.6 - j 0.701	-1.25	GE = 1, GN = 2
Mmana-gal	10.5 - j 17379	1.6	

0.1 m above real gnd			
	Zin (ohms)	Gain (max) dB	
Neomom	38.4 - j 2478	-1.67	
Nec 5	44.6 - j 2794	-2.08	GE = 1, GN = 2
Mmana-gal	10.5 - j 17379	1.62	

Disk Cone

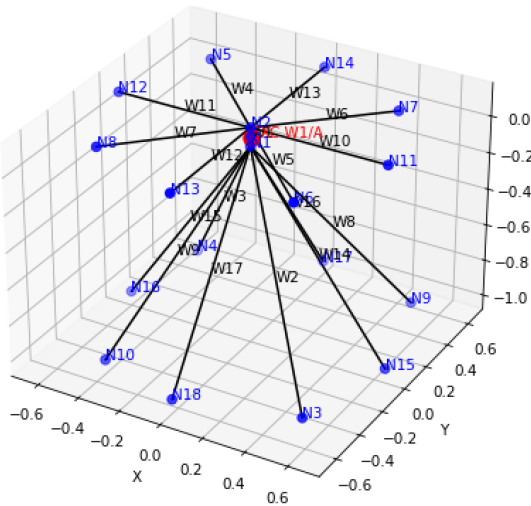
8 Top Hat Radials, 8 Cone Slopers

Discon55-200.maa file in the MMANA-GAL antenna library

f = 145 MHz, wire radius = 5 mm

Free Space

Free Space			
	Zin (ohms)	Gain (max) dB	
Neomom	137.5 + j 35	1.92	$\theta_{\max} = 120 \text{ deg}$
Nec 5	135.6 + j 31.3	1.89	GE = 0
Mmana-gal	134 + j 33.8	1.95	



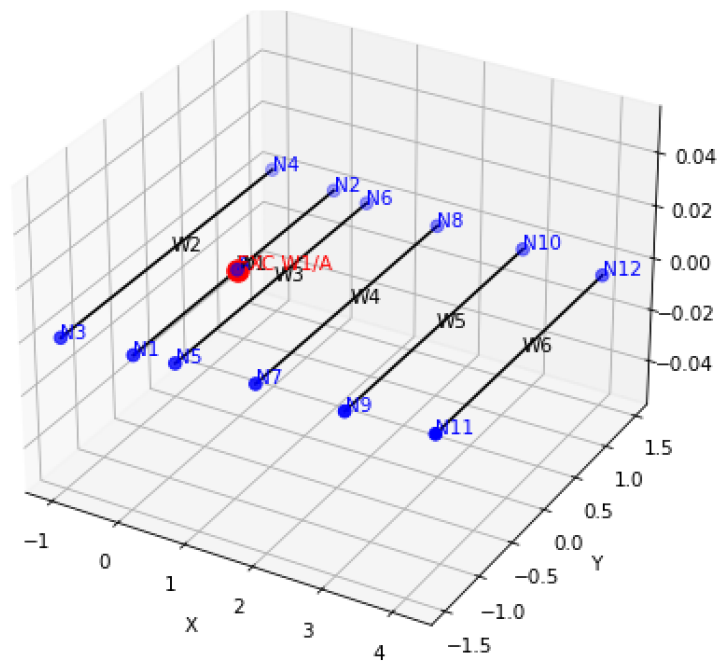
Yagi, Six Element

6elYag.maa file in the MMANA-GAL antenna library

$f = 50.5$ MHz, wire radius = 7 mm

Free Space

Free Space			
	Zin (ohms)	Gain (max) dB	
Neomom	$34.5 - j\ 2.71$	11	
Nec 5	$35.3 - j\ 5.39$	11	GE = 0
Mmana-gal	$34.3 - j\ 2.32$	11	



Delta Loop

$f = 7.25 \text{ MHz}$, wire radius = 1 mm

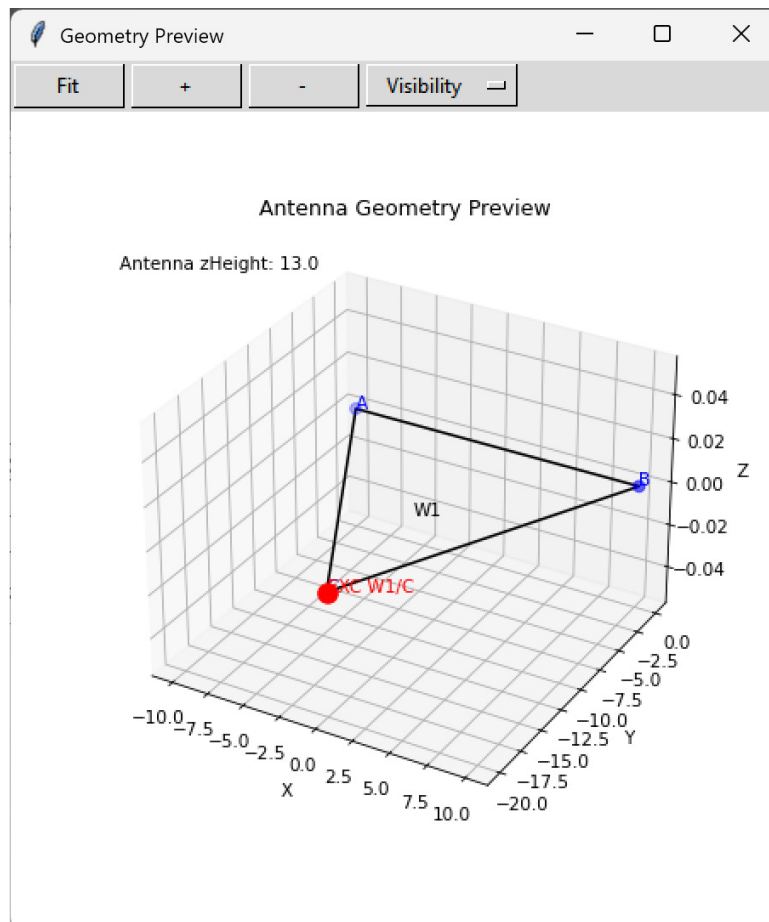
13 m above Real Ground

13m above gnd			
7.25 MHz	Zin (ohms)	Gain (max) dB	
Neomom	1749 - j 1582	5.77	
Nec 5	1945 - j 1563	5.79	
Mmana-gal	1532 - j 1457	5.06	

$f = 29 \text{ MHz}$, wire radius = 1 mm

13 m above Real Ground

13m above gnd			
29 MHz	Zin (ohms)	Gain (max) dB	
Neomom	347 + j 614	13.4	
Nec 5	316 + j 568	13.5	
Mmana-gal	367 + j 637	6.88	

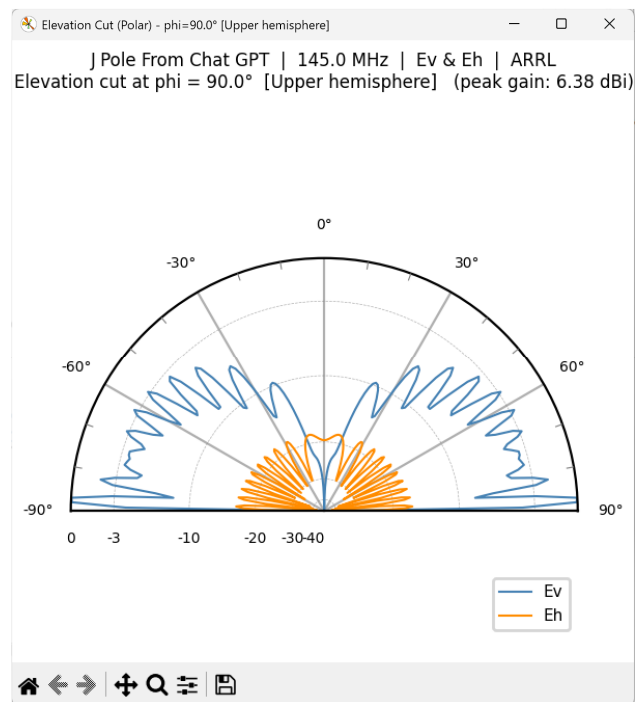
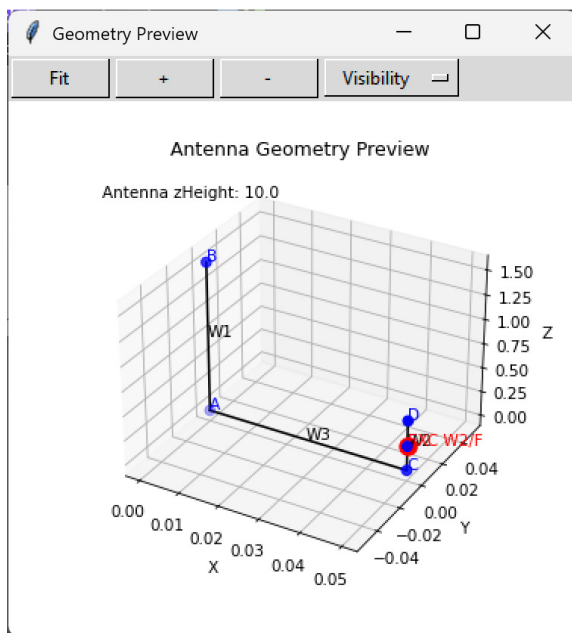


J Pole, 2m

$f = 7.25 \text{ MHz}$, wire radius = 1 mm

10 m above Real Ground

10m above gnd			
145 MHz	Zin (ohms)	Gain (max) dB	
Neomom	$70.2 + j 86.7$	6.38	
Nec 5	$75.1 + j 89.0$	6.56	
Mmana-gal	$72.7 + j 92.2$	6.66	



Neomom *.nml input file example

Hand edit or use neomom_input gui

```
&RunTitle    title = '40 m Delta Loop at 10 m' /

&Frequency_MHz    fmin  = 29.0    fmax  = 29.0    nFreq = 1/

&GROUND    Ground_Plane = real    epsilon = 14.0    sigma = 0.005/

&OPTIONS    NBASISPERLAMBDA = 40/

&node_input    zHeight = 13.0    nNodes = 3    units = meters
    node_list(1) = A, -10.0, 0.0, 0.0
    node_list(2) = B, 10.0, 0.0, 0.0
    node_list(3) = C, 0.0, -20.0, 0.0
/

&wire_primitive
    tag      = W1,
    nNodes   = 4,
    nodeTags = C B A C
    radius   = 0.001
/

&excitation_input
    wireTag = W1
    nodeTag = C
    voltage = 1.0
    phase_deg = 0.0
/
```